Emmanuel G. Maminta

Vision-Language Models · Multimodal & Embodied AI · Physical Retail AI

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PROFESSIONAL EXPERIENCE

OpenText Philippines Jan 2025–Present
AI Developer

- Develop **enterprise AI products and agentic systems**; build LLMOps/MLOps infrastructure and apply parameter-efficient fine-tuning (PEFT) and harness engineering.
- Built an **in-house small language model** to automate proprietary coding tasks for teams in the Philippines and the United States.
- Engineered **knowledge graph-guided retrieval-augmented generation (RAG)** with a unified data model for internal applications.

Bank of the Philippine Islands Nov 2023–Dec 2024
Data Scientist

- Built **personalized product recommendations** for high-net-worth clients.
- Developed unsupervised customer segmentation using the Jobs-to-Be-Done framework to identify distinct market segments and opportunities.

Samsung Research & Development Institute Philippines Apr–Oct 2023
Software Engineer

- Built a proprietary library for **efficient, scalable deep learning experimentation**.

RESEARCH EXPERIENCE

University of the Philippines Diliman Sep 2022–Present
Doctoral Researcher, Ubiquitous Computing Laboratory

- Investigate **3D computer vision for retail**, alongside multimodal and embodied AI.
- Evaluated **190 open-source vision-language models** for zero-shot grocery retrieval, examining pre-training data, architecture, and input resolution (ICPR 2026).
- Led research on an agentic framework for embodied AI in virtual retail; co-authored Sari Sandbox, a benchmark environment with **over 250 interactive grocery items**.
- Investigated open-vocabulary object detection in vision-language models with support from Samsung Research & Development Institute Philippines.

EDUCATION

University of the Philippines Diliman Quezon City, Philippines
Ph.D. in Artificial Intelligence Sep 2022–Present
Expected completion: 2027. Adviser: Prof. Rowel O. Atienza.

M.Eng. in Artificial Intelligence Sep 2022–Jul 2025
B.S. in Electrical Engineering, with honors Aug 2018–Jul 2022

HONORS

Most Active Contributor Award, Microsoft Research Asia Aug 2026
National University of Singapore AI Summer School. Submitted the most contributions to the Global AI Value Challenge during the school.

PUBLICATIONS

A Transferable Variant Signal That Grocery Retrieval Cannot Reach
Emmanuel G. Maminta and Rowel O. Atienza
Asian Conference on Computer Vision (ACCV 2026), ***Main Conference***
Accepted for oral presentation. Osaka, Japan · December 2026.

Late Interaction Ranking for Effective Closed-Catalog Grocery Product Retrieval
Emmanuel G. Maminta and Rowel O. Atienza
European Conference on Computer Vision (ECCV 2026), RetailVision Workshop
Attended and presented. Malmö, Sweden · September 2026.

What Matters for Grocery Product Retrieval with Open Source Vision Language Models
Emmanuel G. Maminta and Rowel O. Atienza
International Conference on Pattern Recognition (ICPR 2026)
Attended and presented. Lyon, France · August 2026. **RRPR reproducibility badge.**

Sari Sandbox: A Virtual Retail Store Environment for Embodied AI Agents
Janika Deborah Gajo, Gerarld Paul Merales, Jerome Escarcha, Brenden Ashley Molina, Gian Nartea, **Emmanuel G. Maminta**, Juan Carlos Roldan, and Rowel O. Atienza
IEEE/CVF International Conference on Computer Vision (ICCV 2025), RetailVision Workshop
Workshop paper. Honolulu, Hawai'i, USA · October 2025.
Featured in Voxel51's Best of ICCV 2025 research event · November 2025.

OPEN-SOURCE SOFTWARE

- **OpenMPR · Initiator.** Started a project sharing grocery retrieval recipes and evaluation resources.
- **stable-worldmodel · Contributor.** Contributed to a reproducible world model research platform.
- **GEPA-Lite · Creator.** Released a lightweight, MIT-licensed Genetic-Pareto prompt optimizer.
- **GEPA · Contributor.** Added AnyMaths for prompt optimization on mathematical word problems.

RESEARCH COMMUNITY

Cohere Labs Open Science Community, Member Sep 2026–Present
Focus: multimodal AI and world models.

CERTIFICATIONS, TRAINING, AND TEACHING

NVIDIA-Certified Professional: Agentic AI
NVIDIA-Certified Associate: Generative AI LLMs

Blender, online course. Currently enrolled.
Oxford Machine Learning Summer School, Jun–Jul 2024. Enrolled in representation learning and generative AI courses.

Teaching, UP Diliman, 2022. Co-taught EEE 111 (Introduction to Programming and Computation) and EEE 121 (Data Structures and Algorithms) with Prof. Jhoanna Rhodette I. Pedrasa.

TECHNICAL SKILLS

Modeling: Python, PyTorch, Transformers (Hugging Face), OpenCLIP, PEFT, OpenCV.
Agents and evaluation: DSPy, LangGraph, LlamaIndex, MLflow, Weights & Biases, Langfuse.
Inference: vLLM, NVIDIA Triton, ONNX, FastAPI.
Infrastructure: Slurm, NVIDIA Jetson, AWS, Azure, Google Cloud.